✨ AI Mood Quest

An AI-Powered Mental Wellness Tracker & Interactive Action Quest

Python • Streamlit • Natural Language Processing • Gamification

BY VIJEY AND HARINIVAAS

AI MOOD QUEST

Project Overview & Vision

🎯 Problem Statement

- Students experience daily fluctuations in emotional well-being (stress, fatigue, excitement).
- Traditional journal apps lack immediate actionable feedback or interactive motivation.
- Students often don't know HOW to reset or improve their mental state during tough school days.

💡 AI Mood Quest Solution

- Analyzes daily diary entries and photos using smart keyword logic.
- Classifies emotional states into 4 distinct categories.
- Delivers instant empathetic AI reflections and actionable coping strategies.
- Gamifies recovery with a 3-step quest, rewards (+50 PTS), and daily streak tracking.

AI MOOD QUEST

Core Application Mechanics

📖 Diary & Photo Input

- Multi-line text area for full diary entry reflections.
- Optional image upload (Pillow integration) for visual snapshots.
- Flexible submission handling.

🧠 4 Mood Categories

- 🔴 Stressed & Overwhelmed
- 🟡 Tired & Low Energy
- ⚡ Excited & Energized
- 🟢 Calm & Balanced
- Tailored 'Why' explanations & reflection.

🎮 Gamification & Timers

- Instant +50 Points reward per completed step.
- Exact 1-day streak increment limit per calendar day.
- Step 3 Running Challenge with Yes/No start prompt.

AI MOOD QUEST

Technical Architecture & Workflow

1. User Input Layer

- Streamlit Web UI
- st.text_area for text
- st.file_uploader for images
- Session state initialization for points & streak

2. Intelligence Engine

- analyze_entry() pipeline
- Keyword evaluation & sentiment mapping
- Generates mood title, 'Why' reasoning, AI response, and 3-step custom quest

3. Interactive Quest UI

- Dynamic Checkbox callbacks
- Stateful Point & Streak calculator (date.today() validation)
- Balloons & Toast notifications upon achievement

AI MOOD QUEST

Mood Classification & Response Logic

🔴 Stressed & Overwhelmed

- Triggers: 'exam', 'test', 'deadline', 'scared', 'fail', 'workload'
- Focus: Micro-steps, deep breathing, running challenge.

🟡 Tired & Low Energy

- Triggers: 'tired', 'sleepy', 'exhausted', 'screen', 'late', 'yawn'
- Focus: Rehydration, physical stretching, sensory refresh.

⚡ Excited & Energized

- Triggers: 'excited', 'happy', 'won', 'awesome', 'hyped', 'win'
- Focus: Channeling motivation into creative goals & physical drives.

🟢 Calm & Balanced

- Triggers: Default steady state / balanced keywords
- Focus: Mindfulness, gratitude reflection, quiet enjoyment.

AI MOOD QUEST

Testing & Demonstration Results

🧪 Tested Diary Scenarios

- Tested with 10+ distinct diary entries including complex school situations.
- Stress Test: 'I have 3 exams tomorrow and I feel overwhelmed!' -> Correctly identified 🔴 Stressed.
- Fatigue Test: 'Stayed up late, brain feels completely drained.' -> Correctly identified 🟡 Tired.

🌟 User Experience Outcomes

- Seamless execution using 'python -m streamlit run app.py'.
- Instant dynamic UI updates without freezing.
- High engagement via visual rewards and clear actionable steps.